16

Asymptotic Optimality[1]

In a rough sense, Theorem 9.14 shows that the maximum likelihood estimator achieves the Cramér–Rao lower bound asymptotically, which suggests that it is asymptotically fully efficient. In this chapter we explore results on asymptotic optimality formalizing notions of asymptotic efficiency and showing that maximum likelihood or similar estimators are efficient in regular cases. Notions of asymptotic efficiency are quite technical and involved, and the treatment here is limited. Our main goal is to derive a result from Hájek (1972), Theorem 16.25 below, which shows that the maximum likelihood estimator is locally asymptotically minimax.

To motivate later results, the first section begins with a curious example that shows why simple approaches in this area fail.

16.1 Superefficiency

Suppose $X_1, X_2, \dots$ are i.i.d. with common density f_θ, $\theta \in \Omega$. By the Cramér–Rao lower bound, if $\delta_n = \delta_n(X_1, \dots, X_n)$ is an unbiased estimator of $g(\theta)$, then

$$\operatorname{Var}_\theta(\delta_n) \geq \frac{\bigl[g'(\theta)\bigr]^2}{nI(\theta)},$$

where

$$I(\theta) = E_\theta \left(\frac{\partial}{\partial\theta} \log f_\theta(X_i) \right)^2$$

is the Fisher information for a single observation. Suppose we drop the assumption that δ_n is unbiased, but assume it is asymptotically normal:

[1] From Section 16.3 on, the results in this chapter are very technical. The material on contiguity in Section 16.2 is needed for the discussion of generalized likelihood ratio tests in Chapter 17, but results from the remaining sections are not used in later chapters.

R.W. Keener, *Theoretical Statistics: Topics for a Core Course*, Springer Texts in Statistics,
DOI 10.1007/978-0-387-93839-4_16,

$$\sqrt{n}\big(\delta_n - g(\theta)\big) \Rightarrow N\big(0, \sigma^2(\theta)\big).$$

This seems to suggest that δ_n is almost unbiased and that

$$\mathrm{Var}_\theta(\sqrt{n}\delta_n) \to \sigma^2(\theta).$$

(This supposition is not automatic, but does hold if the sequence $n\big(\delta_n - g(\theta)\big)^2$ is uniformly integrable.) It seems natural to conjecture that

$$\sigma^2(\theta) \geq \frac{\big[g'(\theta)\big]^2}{I(\theta)}.$$

But the following example, discovered by Hodges in 1951, shows that this conjecture can fail. The import of this counterexample is that a proper formulation of asymptotic optimality will need to consider features of an estimator's distribution beyond the asymptotic variance.

Example 16.1. Let $X_1, X_2, \ldots$ be i.i.d. from $N(\theta, 1)$ and take $\overline{X}_n = (X_1 + \cdots + X_n)/n$. Define δ_n, graphed in Figure 16.1, by

$$\delta_n = \begin{cases} \overline{X}_n, & |\overline{X}_n| \geq 1/n^{1/4}; \\ a\overline{X}_n, & |\overline{X}_n| < 1/n^{1/4}, \end{cases}$$

where a is some constant in $(0, 1)$. Let us compute the limiting distribution of $\sqrt{n}(\delta_n - \theta)$.

Suppose $\theta < 0$. Fix x and consider

$$P_\theta\big(\sqrt{n}(\delta_n - \theta) \leq x\big) = P(\delta_n \leq \theta + x/\sqrt{n}).$$

Since $\theta + x/\sqrt{n} \to \theta < 0$ and $-1/n^{1/4} \to 0$, for n sufficiently large, $\theta + x/\sqrt{n} < -1/n^{1/4}$, and then

$$P_\theta\big(\sqrt{n}(\delta_n - \theta) \leq x\big) = P_\theta(\overline{X}_n \leq \theta + x/\sqrt{n}) = \Phi(x).$$

So in this case, $\sqrt{n}(\delta_n - \theta) \Rightarrow N(0, 1)$. A similar calculation shows that $\sqrt{n}(\delta_n - \theta) \Rightarrow N(0, 1)$ when $\theta > 0$.

Suppose now that $\theta = 0$. Fix x and consider

$$P_\theta(\sqrt{n}\delta_n \leq x) = P_\theta(\delta_n \leq x/\sqrt{n}).$$

For n sufficiently large, $a|x|$ will be less than $n^{1/4}$, or, equivalently, $a|x\sqrt{n}| < 1/n^{1/4}$, and then

$$P_0(\sqrt{n}\delta_n \leq x) = P_0(a\overline{X}_n \leq x/\sqrt{n}) = \Phi(x/a).$$

This is the cumulative distribution function for $N(0, a^2)$. So when $\theta = 0$, $\sqrt{n}(\delta_n - \theta) \Rightarrow N(0, a^2)$.

These calculations show that in general,

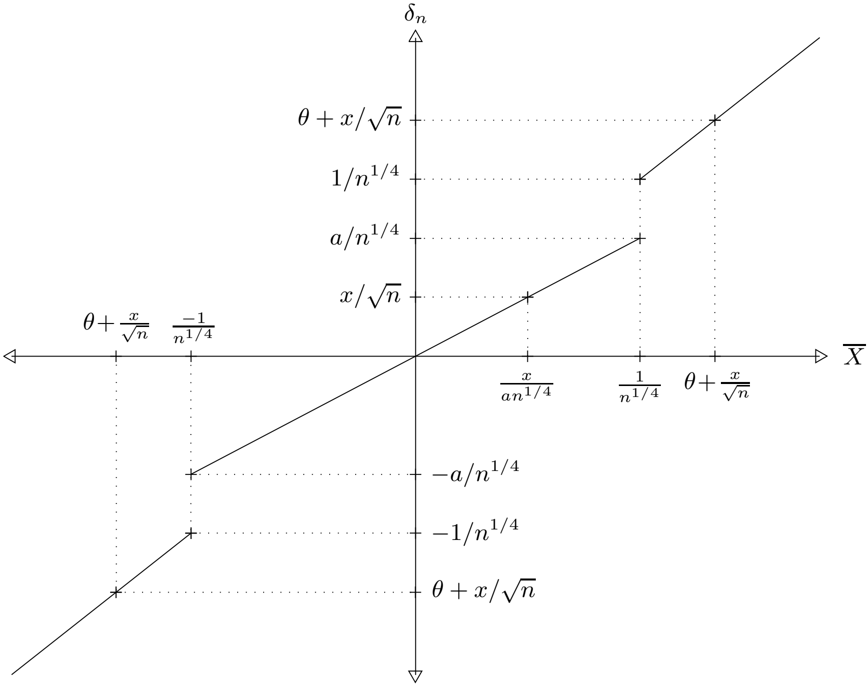

Fig. 16.1. The Hodges' estimator δ_n.

$$\sqrt{n}(\delta_n - \theta) \Rightarrow N\bigl(0, \sigma^2(\theta)\bigr), \tag{16.1}$$

where

$$\sigma^2(\theta) = \begin{cases} 1, & \theta \neq 0; \\ a^2, & \theta = 0. \end{cases}$$

This estimator is called "superefficient" since the variance of the limiting distribution when $\theta = 0$ is smaller than $1/I(\theta) = 1$.

Because $\sqrt{n}(\overline{X}_n - \theta) \sim N(0,1)$, (16.1) seems to suggest that δ_n may be a better estimator than $\overline{X}_n$ when n is large. To explore what is going on, let us consider the risk functions for these estimators under squared error loss. Since $R(\theta, \overline{X}_n) = E_\theta(\overline{X}_n - \theta)^2 = 1/n$, $nR(\theta, \overline{X}_n) = 1$. It can be shown that

$$nR(\theta, \delta_n) \to \begin{cases} 1, & \theta \neq 0; \\ a^2, & \theta = 0, \end{cases}$$

as one might expect from (16.1). But comparison of δ_n and $\overline{X}_n$ by pointwise convergence of their risk functions scaled up by n does not give a complete picture, because the convergence is not uniform in θ. One simple way to see this is to note (see Figure 16.1) that δ_n never takes values in the interval

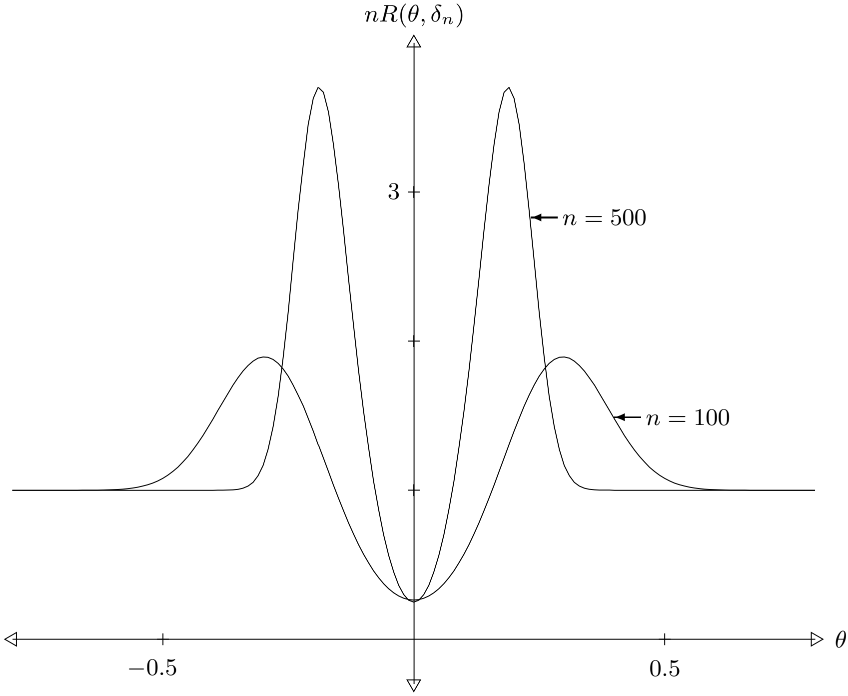

Fig. 16.2. Scaled risk of δ_n with $n = 100$ and $n = 500$ ($a = 1/2$).

$$\left(\frac{a}{n^{1/4}}, \frac{1}{n^{1/4}}\right).$$

If we define

$$\theta_n = \frac{1+a}{2n^{1/4}}$$

to be the midpoint of this interval, then δ_n will always miss θ_n by at least half the width of the interval, and so

$$(\delta_n - \theta_n)^2 \geq \left(\frac{1-a}{2n^{1/4}}\right)^2 = \frac{(1-a)^2}{4\sqrt{n}}.$$

From this,

$$nR(\theta_n, \delta_n) \geq n\frac{(1-a)^2}{4\sqrt{n}} = \frac{(1-a)^2}{4}\sqrt{n} \to \infty,$$

as $n \to \infty$. This shows that for large n the risk of δ_n at θ_n will be much worse than the risk of $\overline{X}_n$ at θ_n. Figure 16.2 plots n times the risk function for δ_n when $n = 100$ and $n = 500$ with a fixed at $1/2$. As n increases, the improved risk near zero does not seem sufficient compensation for the worsening risk nearby.

16.2 Contiguity

Recall that a measure $\tilde{Q}$ is absolutely continuous with respect to another measure Q if $\tilde{Q}(N) = 0$ whenever $Q(N) = 0$. To impart a statistical consequence to this notion, suppose we are interested in testing $H_0 : X \sim Q$ versus $H_1 : X \sim \tilde{Q}$. If the level $\alpha = E_0\varphi(X)$ is zero, then $N = \{x : \varphi(x) > 0\}$ is a null set under Q, and must then also be a null set under $\tilde{Q}$. So the power $\beta = E_1\varphi(X)$ must also be zero. Conversely, $\beta > 0$ implies $\alpha > 0$. If the measures Q and $\tilde{Q}$ are mutually absolutely continuous, $\alpha < 1$ implies $\beta < 1$. In this sense the competing distributions Q and $\tilde{Q}$ are hard to distinguish, at least perfectly.

Contiguity might be viewed as an asymptotic notion of absolute continuity. It concerns two sequences of distributions, $\tilde{Q}_n$ and Q_n, $n = 1, 2, \dots$. These might be viewed as competing joint distributions for data, with n representing the sample size. So $\tilde{Q}_n$ and Q_n are defined on a common measurable space $(\mathcal{X}_n, \mathcal{A}_n)$, but these spaces generally vary with n. For instance, $\mathcal{X}_n$ would be $\mathbb{R}^n$ if n is the sample size and the individual observations are univariate.

Definition 16.2. *The measures $\tilde{Q}_n$ are contiguous to the measures Q_n if $\tilde{Q}_n(A_n) \to 0$ whenever $Q_n(A_n) \to 0$.*

Contiguity can also be framed in the statistical context of simple versus simple testing. Suppose φ_n, $n \geq 1$, are tests of $H_0 : X \sim Q_n$ versus $H_1 : X \sim \tilde{Q}_n$ with levels $\alpha_n = E_0\varphi_n(X)$ and powers $\beta_n = E_1\varphi_n(X)$. If $\tilde{Q}_n$ are contiguous to Q_n and $\alpha_n \to 0$ then $\beta_n \to 0$. If the sequences are mutually contiguous (i.e., Q_n is also contiguous to $\tilde{Q}_n$), then $\beta_n \to 1$ implies $\alpha_n \to 1$. In this sense the competing hypotheses remain hard to distinguish as n increases without bound.

Example 16.3. Suppose

$$Q_n = N_n(\mu_n, \sigma^2 I) \text{ and } \tilde{Q}_n = N_n(\nu_n, \sigma^2 I)$$

where μ_n and ν_n in $\mathbb{R}^n$, $n \geq 1$. By the Neyman–Pearson lemma, the best level α test of $H_0 : X \sim Q_n$ versus $H_1 : X \sim \tilde{Q}_n$ rejects H_0 if

$$(\nu_n - \mu_n) \cdot (X - \mu_n) > \sigma\|\mu_n - \nu_n\|z_\alpha$$

and has power

$$\Phi\left(\frac{\|\mu_n - \nu_n\|}{\sigma} - z_\alpha\right).$$

Suppose $M = \limsup \|\mu_n - \nu_n\| < \infty$, and let A_n, $n \geq 1$, be Borel sets with $A_n \subset \mathbb{R}^n$. The function 1_{A_n}, viewed as a test of H_0 versus H_1 has level $\alpha_n = E_0 1_{A_n} = Q_n(A_n)$ and power $E_1 1_{A_n} = \tilde{Q}_n(A_n)$. But the power of this test is at most that of the optimal test, and so

$$\tilde{Q}_n(A_n) \leq \Phi\left(\frac{\|\mu_n - \nu_n\|}{\sigma} - z_{\alpha_n}\right).$$

But if $Q_n(A_n) = \alpha_n \to 0$, then $z_{\alpha_n} \to \infty$, and from this bound, $\tilde{Q}_n(A_n)$ must also tend to zero. This shows that $\tilde{Q}_n$, $n \geq 1$, are contiguous to Q_n, $n \geq 1$. And because we could reverse the roles of μ_n and ν_n without changing $\|\mu_n - \nu_n\|$, when $\limsup \|\mu_n - \nu_n\| < \infty$ the sequences will be mutually contiguous.

Suppose instead that $\|\mu_n - \nu_n\| \to \infty$. If we take

$$A_n = \Bigl\{x \in \mathbb{R}^n : (\nu_n - \mu_n) \cdot \bigl(x - \tfrac{1}{2}(\mu_n + \nu_n)\bigr) > 0\Bigr\},$$

corresponding to the critical region for a symmetric likelihood ratio test, then

$$Q_n(A_n) = \Phi\left(-\frac{1}{2\sigma}\|\mu_n - \nu_n\|\right) \to 0$$

and

$$\tilde{Q}_n(A_n) = \Phi\left(\frac{1}{2\sigma}\|\mu_n - \nu_n\|\right) \to 1.$$

So in this case the measures are not contiguous. Taking subsequences, they are also not contiguous if $\limsup \|\mu_n - \nu_n\| = \infty$.

If $\mu_n = \theta\mathbf{1}$ and $\nu_n = (\theta + \delta_n)\mathbf{1}$, then under Q_n the entries of X are i.i.d. from $N(\theta, \sigma^2)$, and under $\tilde{Q}_n$ the entries are i.i.d. from $N(\theta + \delta_n, \sigma^2)$. In this case the sequences will be contiguous if $\limsup n\delta_n^2 < \infty$. This may be interpreted as meaning that shifts in the common mean of order $1/\sqrt{n}$ cannot be detected with probability approaching one. This sort of behavior is typical in regular models; see Theorem 16.10 below.

Considering the role of the Neyman–Pearson lemma in this example, it seems natural that contiguity should be related to likelihood ratios. The notion of uniform integrability also plays a role. If X is integrable, then $E|X|I\bigl\{|X| \geq t\bigr\} \to 0$, as $t \to \infty$, by dominated convergence, and our Definition 8.15 of uniform integrability asserts that this holds uniformly over a collection of random variables.

Lemma 16.4. *Suppose $\tilde{Q}_n \ll Q_n$ with L_n the density (or likelihood ratio) $d\tilde{Q}_n/dQ_n$. Let $X_n \sim Q_n$. If the likelihood ratios $L_n(X_n)$, $n \geq 1$, are uniformly integrable, then the measures $\tilde{Q}_n$ are contiguous to the measures Q_n.*

As in Lemma 12.18, the likelihood ratios L_n would usually be computed as $\tilde{p}_n/p_n$, with $\tilde{p}_n$ and p_n the densities of $\tilde{Q}_n$ and Q_n with respect to some measure μ_n.

Proof. Using L_n, and letting E denote expectation when $X_n \sim Q_n$, we have the following bound, valid for any $t > 0$:

$$\begin{aligned}\tilde{Q}_n(A_n) = \int_{A_n} L_n\,dQ_n &= EL_n(X_n)I\{X_n \in A_n\} \\ &\leq tEI\{X_n \in A_n\} + EL_n(X_n)I\{L_n(X_n) \geq t\}.\end{aligned}$$

The first term in this bound is $tQ_n(A_n)$, which tends to zero if $Q_n(A_n) \to 0$. So in this case

$$\limsup \tilde{Q}_n(A_n) \le \limsup EL_n(X_n)I\{L_n(X_n) \ge t\}.$$

Using uniform integrability, the right-hand side here tends to zero as $t \to \infty$, and so $\limsup \tilde{Q}_n(A_n)$ must be zero, proving the lemma. $\square$

The next result shows that convergence in probability remains in effect following a shift to a contiguous sequence of distributions.

Proposition 16.5. *Suppose $X_n \sim Q_n$ and $\tilde{X}_n \sim \tilde{Q}_n$, let T_n be an arbitrary sequence of estimators, and assume $\tilde{Q}_n$, $n \ge 1$, are contiguous to Q_n, $n \ge 1$. If $T_n(X_n) \xrightarrow{p} c$, then $T_n(\tilde{X}_n) \xrightarrow{p} c$. Similarly, if $T_n(X_n) = O_p(1)$, then $T_n(\tilde{X}_n) = O_p(1)$.*

Proof. From the definition of convergence in probability, for any $\epsilon > 0$,

$$P\bigl(|T_n(X_n) - c| \ge \epsilon\bigr) \to 0.$$

Viewing this probability as the Q_n-measure of a set, by contiguity

$$P\bigl(|T_n(\tilde{X}_n) - c| \ge \epsilon\bigr) \to 0,$$

and since ϵ is an arbitrary positive number, $T_n(\tilde{X}_n) \xrightarrow{p} c$. The second assertion can be established with a similar argument. $\square$

To state the final result about contiguity in its proper generality, we again want to view functions as points in a vector space, as we did in Section 12.5, now with a different notion of convergence. Given a measure μ on $(\mathcal{X}, \mathcal{B})$, let

$$\mathcal{L}_2(\mu) = \left\{ f : \int f^2\, d\mu < \infty \right\},$$

and define the $\mathcal{L}_2$-*length* of a function $f \in \mathcal{L}_2(\mu)$ as

$$\|f\|_2 = \left(\int f^2\, d\mu \right)^{1/2}.$$

Then $\|f - g\|_2$ represents the distance between two functions f and g in $\mathcal{L}_2(\mu)$, and with this distance $\mathcal{L}_2(\mu)$ is a metric space,[2] similar in many respects to $\mathbb{R}^n$. Using this distance we have the following natural notions of convergence and differentiation.

[2] To be more precise, $\mathcal{L}_2(\mu)$ is a pseudometric space, because $\|f - g\|$ can be zero for functions $f \ne g$ if they differ only on a null set. It would be a proper metric space if we were to introduce equivalence classes of functions and consider two functions the same if they agreed almost everywhere.

Definition 16.6. *A sequence of functions* $f_n \in \mathcal{L}_2(\mu)$ converges in $\mathcal{L}_2$ *to* $f \in \mathcal{L}_2(\mu)$*, denoted* $f_n \overset{\mathcal{L}_2}{\to} f$*, if* $\|f_n - f\|_2 \to 0$.

Definition 16.7. *A mapping* $\theta \rightsquigarrow f_\theta$ *from* $\mathbb{R}$ *to* $\mathcal{L}_2(\mu)$ *is* differentiable in quadratic mean *at* θ_0 *with derivative* V *if* $V \in \mathcal{L}_2(\mu)$ *and*

$$\frac{f_{\theta_0+\epsilon} - f_{\theta_0}}{\epsilon} \overset{\mathcal{L}_2}{\to} V,$$

as $\epsilon \to 0$.

When the domain of the map is $\mathbb{R}^p$, the derivative, analogous to the gradient in multivariate calculus, will be a vector-valued function. First-order Taylor approximation should give $f_{\theta_0+\epsilon} \approx f_{\theta_0} + \epsilon \cdot V$, which motivates the following definition.

Definition 16.8. *A mapping* $\theta \rightsquigarrow f_\theta$ *from* $\mathbb{R}^p$ *to* $\mathcal{L}_2(\mu)$ *is* differentiable in quadratic mean *at* θ_0 *with derivative* V *if* $\int \|V\|^2 \, d\mu < \infty$ *and*

$$\frac{f_{\theta_0+\epsilon} - f_{\theta_0} - \epsilon \cdot V}{\|\epsilon\|} \overset{\mathcal{L}_2}{\to} 0,$$

as $\epsilon \to 0$.

This notion of differentiation is generally weaker than pointwise differentiability. In most cases the following lemma allows us to compute this derivative as the gradient, provided it exists.

Lemma 16.9. *Let* $\theta \rightsquigarrow f_\theta$ *be a mapping from* $\mathbb{R}^p$ *to* $\mathcal{L}_2(\mu)$. *If* $\nabla_\theta f_\theta(x)$ *exists for almost all* x*, for* θ *in some neighborhood of* θ_0*, and if*

$$\int \|\nabla_\theta f_\theta\|^2 \, d\mu$$

is continuous at θ_0*, then the mapping is differentiable in quadratic mean at* θ_0 *with derivative the gradient* $\nabla_\theta f_\theta$ *evaluated at* $\theta = \theta_0$.

Returning to statistics, let $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ be a dominated family with densities p_θ, $\theta \in \Omega$, and assume for now that $\Omega \subset \mathbb{R}$. Then the functions $\sqrt{p_\theta}$ can be viewed as points in $\mathcal{L}_2(\mu)$. When ordinary derivatives exist, (4.14) and the chain rule give

$$I(\theta) = \int \left(\frac{\partial \log p_\theta}{\partial \theta}\right)^2 p_\theta \, d\mu = 4 \int \left(\frac{\partial \sqrt{p_\theta}}{\partial \theta}\right)^2 \, d\mu.$$

This suggests the following definition for Fisher information:

$$I(\theta) = 4\|V_\theta\|_2^2, \tag{16.2}$$

with V_θ the quadratic mean derivative of $\theta \rightsquigarrow \sqrt{p_\theta}$. This definition is more general and proper than the formulas given earlier that require extra regularity. And the next result shows that the regularity necessary to define Fisher information in this way gives contiguity in i.i.d. models with parameter shifts of order $1/\sqrt{n}$.

Theorem 16.10. *If $\mathcal{P} = \{P_\theta : \theta \in \Omega\}$ is a dominated family with densities p_θ, and if $\theta \rightsquigarrow \sqrt{p}_\theta$ is differentiable in quadratic mean at θ_0, then the measures $P^n_{\theta_0+\Delta/\sqrt{n}}$, $n \ge 1$, are contiguous to measures $P^n_{\theta_0}$, $n \ge 1$.*

16.3 Local Asymptotic Normality

Previous chapters provide a fair amount of information about optimal estimation sampling from a normal distribution with unknown mean. For instance, if $X \sim N(\theta, 1)$, $\theta \in \mathbb{R}$, then X is complete sufficient, and it is the UMVU and best equivariant estimator of θ under squared error loss. And under squared error loss, it is also minimax, minimizing $\sup_\theta E_\theta(\delta - \theta)^2$. This is established and generalized in Section 16.6.

In large samples, the maximum likelihood estimator $\hat\theta$ is approximately normal (after suitable rescaling). If $\hat\theta$ provides most of the information from the data, it would be natural to hope that inference from large samples may be similar to inference sampling from a normal distribution. Naturally, this will involve some rescaling, because with large samples small changes in parameter values will be noticeable from our data. This notion is made precise by considering likelihood ratios; a sequence of distribution families is called *locally asymptotically normal* if the likelihood ratios for the families are close to those for normal distributions, in an appropriate sense.

Suppose $X \sim P_\theta = N_p(\theta, \Sigma)$, $\theta \in \mathbb{R}^p$, with Σ a fixed positive definite covariance matrix. Then the log-likelihood ratio between parameter values t and 0 is

$$\begin{aligned} \ell(t, 0) &= \log \frac{dP_t}{dP_0}(X) \\ &= -\frac{1}{2}(X - t)'\Sigma^{-1}(X - t) + \frac{1}{2}X'\Sigma^{-1}X \\ &= t'\Sigma^{-1}X - \frac{1}{2}t'\Sigma^{-1}t, \end{aligned} \tag{16.3}$$

which is a quadratic function of t with the linear coefficients random and the quadratic coefficients constant.

To see why likelihood ratios are approximately of this form in large samples, suppose X_i, $i \ge 1$, are i.i.d. with common density f_θ, $\theta \in \Omega \subset \mathbb{R}$. Fix θ and define $W_i(\omega) = \log\bigl(f_\omega(X_i)/f_\theta(X_i)\bigr)$. Under sufficient regularity (see Section 4.5),

$$E_\theta W_i'(\theta) = E_\theta \frac{\partial \log f_\theta(X_i)}{\partial \theta} = 0,$$

$$\operatorname{Var}_\theta\bigl(W_i'(\theta)\bigr) = E_\theta \left(\frac{\partial \log f_\theta(X_i)}{\partial \theta}\right)^2 = I(\theta),$$

and

$$E_\theta W_i''(\theta) = E_\theta \frac{\partial^2 \log f_\theta(X_i)}{\partial \theta^2} = -I(\theta).$$

By the central limit theorem,

$$S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n W_i'(\theta) \Rightarrow N\bigl(0, I(\theta)\bigr),$$

and by the law of large numbers,

$$\frac{1}{n} \sum_{i=1}^n W_i''(\theta) \xrightarrow{p} -I(\theta).$$

A two-term Taylor expansion then suggests the following approximation for log-likelihood ratios between θ and "contiguous" alternative $\theta + t/\sqrt{n}$:

$$\begin{aligned}
\ell_n(\theta + t/\sqrt{n}, \theta) &= \log\left[\frac{\prod_{i=1}^n f_{\theta+t/\sqrt{n}}(X_i)}{\prod_{i=1}^n f_\theta(X_i)}\right] \\
&= \sum_{i=1}^n W_i(\theta + t/\sqrt{n}) \\
&\approx \frac{t}{\sqrt{n}} \sum_{i=1}^n W_i'(\theta) + \frac{t^2}{2n} \sum_{i=1}^n W_i''(\theta) \\
&\approx tS_n - \frac{1}{2} t^2 I(\theta).
\end{aligned} \tag{16.4}$$

This is quite similar in form to (16.3)

The following definition formalizes conditions on likelihood ratios sufficient for the notions of asymptotic optimality developed in Section 16.6, yet weak enough to hold in a wide class of applications. These applications include cases where the data are not identically distributed and cases where the data are dependent.

Definition 16.11. *Consider a sequence of models,*

$$\mathcal{P}_n = \{P_{\theta,n} : \theta \in \Omega \subset \mathbb{R}^p\}, \qquad n \geq 1,$$

and let ℓ_n denote log-likelihood ratios for $\mathcal{P}_n$,

$$\ell_n(\omega,\theta) = \log\left[\frac{dP_{\omega,n}}{dP_{\theta,n}}\right].$$

These models are locally asymptotically normal *(LAN) at a parameter value θ in the interior of Ω if there exist random vectors $S_n = S_n(\theta)$ and a positive definite matrix $J = J(\theta)$ such that:*

1. *If t_n is any convergent sequence, $t_n \to t \in \mathbb{R}^p$,*
$$\ell_n(\theta + t_n/\sqrt{n}, \theta) - \left[t'S_n - \tfrac{1}{2}t'Jt\right] \overset{P_{\theta,n}}{\to} 0$$
as $n \to \infty$.
2. *Under $P_{\theta,n}$, $S_n \Rightarrow Z \sim N(0,J)$ as $n \to \infty$.*

Remark 16.12. The second condition in this definition can be replaced by the condition that the measures $P_{\theta,n}$ and $P_{\theta_n,n}$ are contiguous whenever $\sqrt{n}(\theta_n - \theta)$ remains bounded. Mixtures of these measures are also contiguous if the mixing distributions concentrate appropriately near θ. Specifically, if $B(r)$ denotes the ball of radius r about θ and if π_n are probability distributions on Ω such that $\liminf \pi_n\bigl(B(c/\sqrt{n})\bigr) \uparrow 1$ as $c \to \infty$, then $P_{\theta,n}$ and $\int P_{\omega,n}d\pi_n(\omega)$ are contiguous.

Remark 16.13. If the models are LAN and $t_n \to t$, the distributions of S_n under $P_{\theta+t_n/\sqrt{n},n}$ are also approximately normal. Specifically, under $P_{\theta+t_n/\sqrt{n},n}$, $S_n \Rightarrow N(Jt,J)$. To understand the nature of the argument, assume

$$P_{\theta+t_n/\sqrt{n},n} \ll P_{\theta,n},$$

and let f be a bounded continuous function. With suitable uniform integrability, one would then expect

$$\begin{aligned} E_{\theta+t_n/\sqrt{n},n} f(S_n) &= E_{\theta,n} f(S_n) e^{\ell_n(\theta+t_n/\sqrt{n},\theta)} \\ &\approx Ef(Z)e^{t'Z - t'Jt/2} = Ef(Jt+Z). \end{aligned}$$

Theorem 16.14. *Suppose $X_1, X_2, \dots$ are i.i.d. with common density p_θ, and let $P_{\theta,n}$ be the joint distribution of $X_1,\dots,X_n$. If the mapping $\omega \rightsquigarrow \sqrt{p_\omega}$ is differentiable in quadratic mean at a parameter value θ in the interior of the parameter space $\Omega \subset \mathbb{R}^p$, then the families $\mathcal{P}_n = \{P_{\theta,n} : \theta \in \Omega\}$ are locally asymptotically normal at θ with J the Fisher information given in* (16.2), *$J = I(\theta)$.*

The quadratic approximation $t'S_n - \frac{1}{2}t'Jt$ in the LAN definition is maximized at $\hat{t} = J^{-1}S_n$. Because the maximum likelihood estimator $\hat{\theta}_n$ maximizes

$$l_n(\omega) - l_n(\theta) = \ell_n\left(\theta + \frac{\sqrt{n}(\omega-\theta)}{\sqrt{n}}, \theta\right)$$

over $\omega \in \Omega$, the LAN approximation suggests that

$$\sqrt{n}(\hat{\theta}_n - \theta) \approx J^{-1} S_n.$$

Regularity conditions akin to those of Theorem 9.14 but extended to the multivariate case ensure that

$$\sqrt{n}(\hat{\theta}_n - \theta) - J^{-1} S_n \xrightarrow{P_\theta} 0, \tag{16.5}$$

as $n \to \infty$. We assume as we proceed that (16.5) holds for suitable estimators $\hat{\theta}_n$, but these estimators need not be maximum likelihood.

Example 16.15. Suppose $X_1, X_2, \dots$ are i.i.d. absolutely continuous random vectors in $\mathbb{R}^4$ with common density

$$p_\theta(x) = c \frac{e^{-\|x-\theta\|^2}}{\|x - \theta\|}.$$

The families of joint distributions here are LAN. With the pole in the density, the likelihood function is infinite at each data point, so a maximum likelihood estimator will be one of the observed data. Section 6.3 of Le Cam and Yang (2000) details a general method to find estimators $\hat{\theta}_n$ satisfying (16.5). This method is based on using the LAN approximation to improve a reasonable preliminary estimator, such as $\overline{X}_n$ in this example.

16.4 Minimax Estimation of a Normal Mean

An estimator δ is called *minimax* if it minimizes $\sup_{\theta \in \Omega} R(\theta, \delta)$. In this section we find minimax estimates for the mean of a normal distribution. These results are used in the next section when a locally asymptotically minimax notion of asymptotic optimality is developed.

As an initial problem, suppose $X \sim N_p(\theta, I)$, $\theta \in \mathbb{R}^p$, and consider a Bayesian model with prior distribution

$$\Theta \sim N(0, \sigma^2 I).$$

Then

$$\Theta | X = x \sim N\left(\frac{\sigma^2 x}{1+\sigma^2}, \frac{\sigma^2}{1+\sigma^2} I\right),$$

and the Bayes estimator under compound squared error loss is

$$\tilde{\theta} = \frac{\sigma^2 X}{1+\sigma^2},$$

with Bayes risk

$$E\|\tilde{\theta} - \Theta\|^2 = EE\big[\|\tilde{\theta} - \Theta\|^2 \bigm| X\big] = \frac{p\sigma^2}{1+\sigma^2}.$$

Because $\tilde{\theta}$ is Bayes, conditioning on Θ gives

$$\begin{aligned} E\|\tilde{\theta}-\Theta\|^2 &= EE\big[\|\tilde{\theta}-\Theta\|^2 \mid \Theta\big] \\ &= ER(\Theta,\tilde{\theta}) \le ER(\Theta,\delta) \le \sup_\theta R(\theta,\delta), \end{aligned}$$

for any competing estimator δ. So for any δ,

$$\sup_\theta R(\theta,\delta) \ge p\frac{\sigma^2}{1+\sigma^2}. \tag{16.6}$$

But this holds for any σ, so letting $\sigma \to \infty$ we have

$$\sup_\theta R(\theta,\delta) \ge p$$

for any δ. But the risk of $\delta(X) = X$ equals p, and so X is minimax.

As an extension, we show that X is also minimax when the loss function has form $L(\theta, d) = W(d-\theta)$ with W "bowl-shaped" according to the following definition.

Definition 16.16. *A function $W : \mathbb{R}^p \to [0,\infty]$ is* bowl-shaped *if $\{x : W(x) \le \alpha\}$ is convex and symmetric about zero for every $\alpha \ge 0$.*

The following result due to Anderson (1955) is used to find Bayes estimators with bowl-shaped loss functions.

Theorem 16.17 (Anderson's lemma). *If f is a Lebesgue density on $\mathbb{R}^p$ with $\{x : f(x) \ge \alpha\}$ convex and symmetric about zero for every $\alpha \ge 0$, and if W is bowl-shaped, then*

$$\int W(x-c)f(x)\,dx \ge \int W(x)f(x)\,dx,$$

for every $c \in \mathbb{R}^p$.

The proof of this result relies on the following inequality.

Theorem 16.18 (Brunn–Minkowski). *If A and B are nonempty Borel sets in $\mathbb{R}^p$ with sum $A+B = \{x+y : x \in A, y \in B\}$ (the Minkowski sum of A and B), and λ denotes Lebesgue measure, then*

$$\lambda(A+B)^{1/p} \ge \lambda(A)^{1/p} + \lambda(B)^{1/p}.$$

Proof. Let a *box* denote a bounded Cartesian product of intervals and suppose A and B are both boxes with $a_1,\dots,a_p$ the lengths of the sides of A and $b_1,\dots,b_p$ the lengths of the sides of B. Then

$$\lambda(A) = \prod_{i=1}^p a_i \text{ and } \lambda(B) = \prod_{i=1}^p b_i.$$

The sum $A + B$ is also a box, and the lengths of the sides of this box are $a_1 + b_1, \ldots, a_p + b_p$. Thus

$$\lambda(A + B) = \prod_{i=1}^{p}(a_i + b_i).$$

Since arithmetic averages bound geometric averages (see Problem 3.32),

$$\left(\prod_{i=1}^{p} \frac{a_i}{a_i + b_i}\right)^{1/p} + \left(\prod_{i=1}^{p} \frac{b_i}{a_i + b_i}\right)^{1/p} \le \frac{1}{p}\sum_{i=1}^{p} \frac{a_i}{a_i + b_i} + \frac{1}{p}\sum_{i=1}^{p} \frac{b_i}{a_i + b_i} = 1,$$

which gives the desired inequality for boxes.

We next show that the inequality holds when A and B are both finite unions of disjoint boxes. The proof is based on induction on the total number of boxes in A and B, and there is no harm assuming that A has at least two boxes (if not, just switch A and B). Translating A (if necessary) we can assume that some coordinate hyperplane, $\{x : x_k = 0\}$ separates two of the boxes in A. Define half-spaces $H_+ = \{x : x_k \ge 0\}$, $H_- = \{x : x_k < 0\}$ and let $A_\pm$ be intersections of A with these half spaces, $A_+ = A \cap H_+$ and $A_- = A \cap H_-$. Note that $A_\pm$ are both finite intersections of boxes with the total number of boxes in each of them less than the number of boxes in A. The proportion of the volume of A in H_+ is $\lambda(A_+)/\lambda(A)$, and by translating B we make $\lambda(B \cap H_+)/\lambda(B)$ match this proportion. Defining $B_\pm = B \cap H_\pm$ we then have

$$\frac{\lambda(A_\pm)}{\lambda(A)} = \frac{\lambda(B_\pm)}{\lambda(B)}. \tag{16.7}$$

Because intersection with a half-plane cannot increase the number of boxes in a set, the number of boxes in A_+ and B_+ and the number of boxes in A_- and B_- are both less than the number of boxes in A and B, and by the inductive hypothesis we can assume that the inequality holds for both of these pairs. Also note that since $A_+ + B_+ \subset H_+$ and $A_- + B_- \subset H_-$,

$$\lambda\bigl((A_+ + B_+) \cup (A_- + B_-)\bigr) = \lambda(A_+ + B_+) + \lambda(A_- + B_-),$$

and that $A + B \supset (A_+ + B_+) \cup (A_- + B_-)$. Using these, (16.7), and the inductive hypothesis,

$$\begin{aligned}
\lambda(A + B) &\ge \lambda(A_+ + B_+) + \lambda(A_- + B_-) \\
&\ge \bigl(\lambda(A_+)^{1/p} + \lambda(B_+)^{1/p}\bigr)^p + \bigl(\lambda(A_-)^{1/p} + \lambda(B_-)^{1/p}\bigr)^p \\
&= \lambda(A_+)\left(1 + \frac{\lambda(B)^{1/p}}{\lambda(A)^{1/p}}\right)^p + \lambda(A_-)\left(1 + \frac{\lambda(B)^{1/p}}{\lambda(A)^{1/p}}\right)^p \\
&= \lambda(A)\left(1 + \frac{\lambda(B)^{1/p}}{\lambda(A)^{1/p}}\right)^p \\
&= \bigl(\lambda(A)^{1/p} + \lambda(B)^{1/p}\bigr)^p.
\end{aligned}$$

This proves the theorem when A and B are finite unions of boxes. The general case follows by an approximation argument. As a starting point, we use the fact that Lebesgue measure is *regular*, which means that $\lambda(K) < \infty$ for all compact K, and for any B,

$$\lambda(B) = \inf\{\lambda(V) : V \supset B, V \text{ open}\}$$

and

$$\lambda(B) = \sup\{\lambda(K) : K \subset B, K \text{ compact}\}.$$

Suppose A is open. Fix $\epsilon > 0$ and let $K \subset A$ be a compact set with $\lambda(K) \geq \lambda(A) - \epsilon$. Because the distance between K and A^c is positive, we can cover K with open boxes centered at all points of K so that each box in the cover lies in A. The union $\tilde{A}$ of a finite subcover will then contain K and lie in A, so $\lambda(\tilde{A}) \geq \lambda(A) - \epsilon$, and $\tilde{A}$ will be a finite union of disjoint boxes. Similarly, if B is open there is a set $\tilde{B} \subset B$ that is a finite union of disjoint boxes with $\lambda(\tilde{B}) \geq \lambda(B) - \epsilon$. Because $A + B \supset \tilde{A} + \tilde{B}$,

$$\begin{aligned}\lambda(A+B)^{1/p} \geq \lambda(\tilde{A}+\tilde{B})^{1/p} &\geq \lambda(\tilde{A})^{1/p} + \lambda(\tilde{B})^{1/p} \\ &\geq \bigl(\lambda(A) - \epsilon\bigr)^{1/p} + \bigl(\lambda(B) - \epsilon\bigr)^{1/p}. \end{aligned} \tag{16.8}$$

Letting $\epsilon \to 0$, the inequality holds for nonempty open sets. Next, suppose A and B are both compact. Define open sets

$$A_n = \{x : \|x - y\| < 1/n, \exists y \in A\}$$

and

$$B_n = \{x : \|x - y\| < 1/n, \exists y \in B\}.$$

Then

$$A + B = \bigcap_{n \geq 1} (A_n + B_n),$$

for if s lies in the intersection, then $s = a_n + b_n$ with $a_n \in A_n$ and $b_n \in B_n$, and along a subsequence $(a_n, b_n) \to (a, b) \in A \times B$. Then $s = a + b \in A + B$. Using this,

$$\begin{aligned}\lambda(A+B)^{1/p} &= \lim_{n\to\infty} \lambda(A_n + B_n)^{1/p} \\ &\geq \lim_{n\to\infty} \bigl(\lambda(A_n)^{1/p} + \lambda(B_n)^{1/p}\bigr) \\ &= \lambda(A)^{1/p} + \lambda(B)^{1/p}.\end{aligned}$$

Finally, if A and B are arbitrary Borel sets with positive and finite measure, and if $\epsilon > 0$, there are compact subsets $\tilde{A} \subset A$ and $\tilde{B} \subset B$ such that $\lambda(\tilde{A}) \geq \lambda(A) - \epsilon$ and $\lambda(\tilde{B}) \geq \lambda(B) - \epsilon$. The inequality then follows by the argument used in (16.8) □

Corollary 16.19. *If A and B are symmetric convex subsets of $\mathbb{R}^p$ and c is any vector in $\mathbb{R}^p$, then*

$$\lambda\bigl((c+A)\cap B\bigr) \le \lambda(A\cap B).$$

Proof. Let $K_+ = (c+A)\cap B$ and $K_- = (-c+A)\cap B$. By symmetry $K_- = -K_+$ and so $\lambda(K_+) = \lambda(K_-)$. Define $K = \frac{1}{2}(K_+ + K_-)$, and note that $K \subset A\cap B$. By the Brunn–Minkowski inequality,

$$\begin{aligned}\lambda(K)^{1/p} &\ge \lambda\left(\frac{1}{2}K_+\right)^{1/p} + \lambda\left(\frac{1}{2}K_-\right)^{1/p} \\ &= \frac{1}{2}\lambda(K_+)^{1/p} + \frac{1}{2}\lambda(K_-)^{1/p} = \lambda(K_+)^{1/p}.\end{aligned}$$

So

$$\lambda(A\cap B) \ge \lambda(K) \ge \lambda(K_+) = \lambda\bigl((c+A)\cap B\bigr). \qquad \square$$

Proof of Theorem 16.17. For $u \ge 0$ define convex symmetric sets $A_u = \{x : W(x) \le u\}$ and $B_u = \{x : f(x) \ge u\}$. Using Fubini's theorem and Corollary 16.19,

$$\begin{aligned}\int W(x-c)f(x)\,dx &= \iint_0^\infty \int_0^\infty I\bigl[W(x-c) > u, f(x) \ge v\bigr]\,du\,dv\,dx \\ &= \int_0^\infty \int_0^\infty \int I(x \notin c + A_u, x \in B_v)\,dx\,du\,dv \\ &= \int_0^\infty \int_0^\infty \bigl[\lambda(B_v) - \lambda\bigl((c+A_u)\cap B_v\bigr)\bigr]\,du\,dv \\ &\ge \int_0^\infty \int_0^\infty \bigl[\lambda(B_v) - \lambda(A_u\cap B_v)\bigr]\,du\,dv \\ &= \int W(x)f(x)\,dx.\end{aligned} \qquad \square$$

Theorem 16.20. *Suppose $X \sim N_p(\theta, \Sigma)$ with Σ a known positive definite matrix, and consider estimating the mean θ with loss function $L(\theta, d) = W(\theta - d)$ and W bowl-shaped. Then X is minimax.*

Proof. Consider a Bayesian formulation in which the prior distribution for Θ is $N(0, \sigma^2\Sigma)$. Let $\tilde\Theta = \Sigma^{-1/2}\Theta$ and $\tilde X = \Sigma^{-1/2}X$ and note that

$$\tilde\Theta \sim N(0, \sigma^2 I)$$

and

$$\tilde X | \tilde\Theta = \tilde\theta \sim N(\tilde\theta, I).$$

As before,

$$\tilde{\Theta}|\tilde{X} = \tilde{x} \sim N\left(\frac{\sigma^2 \tilde{x}}{1+\sigma^2}, \frac{\sigma^2}{1+\sigma^2} I\right).$$

Since conditioning on $\tilde{X}$ is the same as conditioning on X, multiplication by $\Sigma^{1/2}$ gives

$$\Theta|X = x \sim N\left(\frac{\sigma^2 x}{1+\sigma^2}, \frac{\sigma^2}{1+\sigma^2} \Sigma\right).$$

If $Z \sim N\big((0, \sigma^2\Sigma/(1+\sigma^2)\big)$, with density f, then the posterior risk of an estimator δ is

$$\begin{aligned} E\big[W\big(\Theta - \delta(X)\big) \mid X = x\big] &= EW\left(Z + \frac{\sigma^2 x}{1+\sigma^2} - \delta(x)\right) \\ &= \int W\left(z + \frac{\sigma^2 x}{1+\sigma^2} - \delta(x)\right) f(z)\,dz. \end{aligned}$$

By Theorem 16.17, this is minimized if $\delta(x) = \sigma^2 x/(1+\sigma^2)$, and so again the Bayes estimator is

$$\tilde{\theta} = \frac{\sigma^2 X}{1+\sigma^2}.$$

If $\epsilon = X - \Theta$, then

$$\epsilon|\Theta = \theta \sim N(0, \Sigma),$$

and so ϵ and Θ are independent and the marginal distribution of ϵ is $N(0, \Sigma^2)$. Using this, the Bayes risk is

$$\begin{aligned} EW(\Theta - \tilde{\theta}) &= EW\left(\Theta - \frac{\sigma^2(\Theta + \epsilon)}{1+\sigma^2}\right) \\ &= EW\left(\frac{\Theta}{1+\sigma^2} - \frac{\sigma^2 \epsilon}{1+\sigma^2}\right) = EW\left(\frac{\sigma\epsilon}{\sqrt{1+\sigma^2}}\right), \end{aligned}$$

which converges to $EW(\epsilon)$ as $\sigma \to \infty$ by monotone convergence. Arguing as we did for (16.6), for any δ

$$\sup_\theta R(\theta, \delta) \geq EW(\epsilon).$$

Since this is the risk of X (for any θ), X is minimax. ☐

16.5 Posterior Distributions

In this section we derive normal approximations for posterior distributions for LAN families. Local asymptotic optimality is derived using these approximations with arguments similar to those in the preceding section.

The approximations for posterior distributions are developed using a notion of convergence stronger than convergence in distribution, based on the following norm.

Definition 16.21. *The* total variation *norm of the difference between two probability measures P and Q is defined as*

$$\|P-Q\| = \sup\bigl\{|\textstyle\int f\,dP - \int f\,dQ| : |f| \le 1\bigr\}.$$

If P and Q have densities p and q with respect to a measure μ, and $|f| \le 1$, then

$$\left|\int f\,dP - \int f\,dQ\right| = \left|\int f(p-q)\,d\mu\right| \le \int |p-q|\,d\mu.$$

This bound is achieved when $f = \operatorname{Sign}(p-q)$, and so

$$\|P-Q\| = \int |p-q|\,d\mu.$$

Taking advantage of the fact that p and q both integrate to one,

$$\begin{aligned}\|P-Q\| &= \int_{p>q}(p-q)\,d\mu + \int_{q>p}(q-p)\,d\mu = 2\int_{p>q}(p-q)\,d\mu \\ &= 2\int_{p>q}(1-L)\,dP = 2\int\bigl(1-\min\{1,L\}\bigr)\,dP, \end{aligned} \tag{16.9}$$

where $L = q/p$ is the likelihood ratio dQ/dP.

If f is a bounded function, $\sup|f| = M$, and we take $f^* = f/M$, then $|f^*| \le 1$ and so

$$\left|\int f\,dP - \int f\,dQ\right| = M\left|\int f^*\,dP - \int f^*\,dQ\right| \le M\|P-Q\|. \tag{16.10}$$

Strong convergence is defined using the total variation norm. Distributions P_n *converge strongly* to P if $\|P_n - P\| \to 0$. If this happens, then by (16.10) $\int f\,dP_n \to \int f\,dP$ for any bounded measurable f. This can be compared with convergence in distribution where, by Theorem 8.9, $\int f\,dP_n \to \int f\,dP$ for bounded *continuous* functions, but convergence can fail if f is discontinuous. So strong convergence implies convergence in distribution.

Lemma 16.22. *Let $\tilde P$ and P be two possible joint distributions for random vectors X and Y. Suppose $\tilde P \ll P$, and let f denote the density $d\tilde P/dP$. Let $\tilde Q$ and Q denote the marginal distributions for X when $(X,Y) \sim \tilde P$ and $(X,Y) \sim P$, and let $\tilde R_x$ and R_x denote the conditional distributions for Y given $X = x$ when $(X,Y) \sim \tilde P$ and $(X,Y) \sim P$. Then $\tilde Q \ll Q$ with density*

$$h(x) = \frac{d\tilde Q}{dQ}(x) = \int f(x,y)\,dR_x(y),$$

and $\tilde R_x \ll R_x$ (a.e.) with density

$$\frac{d\tilde R_x}{dR_x}(y) = \frac{f(x,y)}{h(x)}.$$

Proof. Using Lemma 12.18 and smoothing,

$$\tilde{E}g(X) = Eg(X)f(X,Y) = Eg(X)E[f(X,Y)|X] = \int g(x)h(x)\,dQ(x).$$

To show that the stated densities for the conditional distributions are correct we need to show that iterated integration gives the integral against $\tilde{P}$. This is the case because

$$\begin{aligned}\iint g(x,y)\frac{f(x,y)}{h(x)}\,dR_x(y)\,d\tilde{Q}(x) &= \iint g(x,y)f(x,y)\,dR_x(y)dQ(x)\\ &= EE[g(X,Y)f(X,Y)|X]\\ &= Eg(X,Y)f(X,Y) = \tilde{E}g(X,Y).\end{aligned}$$ ☐

To motivate the main result, approximating posterior distributions, suppose our family is LAN at θ_0, and that the prior distribution for Θ is $N(\theta_0, \Gamma^{-1}/n)$.[3] If the LAN approximation and the approximation (16.5) for $\hat{\theta}$ were exact, then the likelihood function would be proportional to

$$\exp\Bigl[n(\theta-\theta_0)'J(\hat{\theta}_n-\theta_0) - \frac{n}{2}(\theta-\theta_0)'J(\theta-\theta_0)\Bigr],$$

and the posterior distribution for Θ would be

$$G_{x,n} = N\bigl(\theta_0 + (\Gamma+J)^{-1}J(\hat{\theta}_n-\theta_0), (\Gamma+J)^{-1}/n\bigr). \tag{16.11}$$

For convenience, as we proceed dependence on n is suppressed from the notation.

Theorem 16.23. *Suppose our families are LAN at θ_0 in the interior of Ω and that $\hat{\theta}$ satisfies* (16.5). *Consider a sequence of Bayesian models in which $\Theta \sim N(\theta_0, \Gamma^{-1}/n)$ (truncated to Ω) with Γ a fixed positive definite matrix. Let F_x denote the conditional distribution of Θ given $X = x$, and let G_x denote the normal approximation for this distribution given in* (16.11). *Then $\|F_X - G_X\|$ converges to zero in probability as $n \to \infty$.*

Proof. (Sketch) Let P denote the joint distribution of X and Θ (in the Bayesian model), and let Q denote the marginal distribution for X. Introduce another model in which X has the same marginal distribution and $\tilde{\Theta}|X = x \sim G_x$, the normal approximation for the posterior. Let $\tilde{P}$ denote the joint distribution for X and $\tilde{\Theta}$. Finally, let $\overline{P} = \frac{1}{2}(P + \tilde{P})$, so that $P \ll \overline{P}$ and $\tilde{P} \ll \overline{P}$, and introduce densities

$$f(x,\theta) = \frac{dP}{d\overline{P}}(x,\theta) \text{ and } \tilde{f}(x,\theta) = \frac{d\tilde{P}}{d\overline{P}}(x,\theta).$$

[3] If Ω is not all $\mathbb{R}^p$, then we should truncate the prior to Ω. If θ_0 lies in the interior of Ω, only minor changes result and the theorem is correct as stated. But to keep the presentation accessible we do not worry about this issue in the proof.

The marginal distributions for X are the same under P, $\tilde{P}$, and (thus) $\overline{P}$. So both marginal densities for X must be one, and by Lemma 16.22, $f(x,\cdot)$ and $\tilde{f}(x,\cdot)$ are densities for F_x and G_x. Using (16.9),

$$\|F_x - G_x\| = 2\int \bigl[1 - \min\{1, L(x,\theta)\}\bigr]\,dF_x(\theta),$$

where L is the likelihood ratio

$$L(x,\theta) = \frac{\tilde{f}(x,\theta)}{f(x,\theta)}.$$

Integrating against the marginal distribution of X,

$$E\|F_X - G_X\| = 2E\bigl[1 - \min\{1, L(X,\Theta)\}\bigr],$$

and the theorem will follow if $L(X,\Theta) \xrightarrow{p} 1$.

We next want to rewrite L to take advantage of the things we know about the likelihood and G_x. Suppose $P \ll \tilde{P}$. Then P has density $f(x,\theta)/\tilde{f}(x,\theta)$ with respect to $\tilde{P}$. Because the marginal distributions of X are the same under P and $\tilde{P}$, the marginal density must be one, and the formula in Lemma 16.22 then gives

$$\int \frac{f(x,\tilde{\theta})}{\tilde{f}(x,\tilde{\theta})}\,dG_x(\tilde{\theta}) = 1. \tag{16.12}$$

When $P \ll \tilde{P}$ fails, this need not hold exactly, but remains approximately true.[4] Assuming, for convenience, that (16.12) holds exactly, we have

$$L(x,\theta) = \int \frac{\tilde{f}(x,\theta)}{\tilde{f}(x,\tilde{\theta})}\frac{f(x,\tilde{\theta})}{f(x,\theta)}\,dG_x(\tilde{\theta}),$$

and from this the theorem will follow if

$$\frac{\tilde{f}(X,\Theta)}{\tilde{f}(X,\tilde{\Theta})}\frac{f(X,\tilde{\Theta})}{f(X,\Theta)} \xrightarrow{p} 1. \tag{16.13}$$

The two fractions here can both be viewed as likelihood ratios, since f and $\tilde{f}$ are both joint densities. Specifically, viewing $\tilde{f}$ as proportional to a density for X times the normal conditional density G_x,

$$\begin{aligned}\frac{\tilde{f}(X,\Theta)}{\tilde{f}(X,\tilde{\Theta})} = \exp\Bigl[&-\frac{n}{2}(\Theta-\theta_0)'(\Gamma+J)(\Theta-\theta_0)\\ &+\frac{n}{2}(\tilde{\Theta}-\theta_0)'(\Gamma+J)(\tilde{\Theta}-\theta_0)\\ &+n(\hat{\theta}-\theta_0)'J(\Theta-\tilde{\Theta})\Bigr],\end{aligned}$$

---

[4] If H is the conditional distribution given X under $\overline{P}$, then $\int f(x,\tilde{\theta})\,dH_x(\tilde{\theta}) = 1$. Since $dG_x(\tilde{\theta}) = \tilde{f}(x,\tilde{\theta})dH_x(\tilde{\theta})$, the true value for the integral is $1 - P\bigl(\tilde{f}(X,\Theta) = 0 \bigm| X = x\bigr)$. This approaches one by a contiguity argument.

and viewing f as proportional to the normal density for Θ times a conditional density for X given θ,

$$\frac{f(X,\tilde\Theta)}{f(X,\Theta)} = \exp\Big[-\frac{n}{2}(\tilde\Theta-\theta_0)'\Gamma(\tilde\Theta-\theta_0)+\frac{n}{2}(\Theta-\theta_0)'\Gamma(\Theta-\theta_0) + \ell_n(\tilde\Theta,\theta_0)-\ell_n(\Theta,\theta_0)\Big].$$

If the LAN approximation for ℓ_n and approximation (16.5) held exactly, then the left-hand side of (16.13) would be one. The proof is completed by arguing that the approximations imply convergence in probability. □

16.6 Locally Asymptotically Minimax Estimation

Our first lemma uses the approximations of the previous section and Anderson's lemma (Theorem 16.17) to approximate Bayes risks with the loss function a bounded bowl-shaped function.

Lemma 16.24. *Suppose our families are LAN at θ in the interior of Ω and that $\hat\theta$ satisfies* (16.5). *Consider Bayesian models in which the prior distribution for Θ is $N(\theta,\sigma^2J^{-1}/n)$. If W is a bounded bowl-shaped function, then*

$$\lim_{n\to\infty}\inf_\delta E_nW\big(\sqrt{n}(\delta-\Theta)\big) = EW\left(\frac{\sigma Z}{\sqrt{1+\sigma^2}}\right),$$

where $Z\sim N(0,J^{-1})$.

Proof. Let

$$G_{x,n} = N\left(\theta+\frac{\sigma^2(\hat\theta_n-\theta)}{1+\sigma^2},\frac{\sigma^2J^{-1}}{n(1+\sigma^2)}\right),$$

the approximation for the posterior distribution from Theorem 16.23, and let $F_{n,x}$ denote the true posterior distribution of θ. Define

$$\begin{aligned}\rho_n(x) &= \inf_d E_n\big[W\big(\sqrt{n}(d-\Theta)\big)\;\big|\;X=x\big]\\ &= \inf_d\int W\big(\sqrt{n}(d-\theta)\big)\,dF_{x,n}(\theta).\end{aligned}$$

Then, as in Theorem 7.1,

$$\inf_\delta E_nW\big(\sqrt{n}(\delta-\Theta)\big) = E_n\rho_n(X).$$

Using (16.10), for any d,

$$\left|\int W\big(\sqrt{n}(d-\theta)\big)\,dF_{x,n}(\theta) - \int W\big(\sqrt{n}(d-\theta)\big)\,dG_{x,n}(\theta)\right| \\ \leq M\|F_{x,n}-G_{x,n}\|,$$

and it then follows that

$$\left|\rho_n(x) - \inf_d \int W\big(\sqrt{n}(d-\theta)\big)\,dG_{x,n}(\theta)\right| \leq M\|F_{x,n}-G_{x,n}\|,$$

where $M=\sup|W|$. But by Anderson's lemma (Theorem 16.17),

$$\inf_d \int W\big(\sqrt{n}(d-\theta)\big)\,dG_{x,n}(\theta) = EW\left(\frac{\sigma Z}{\sqrt{1+\sigma^2}}\right).$$

So

$$\left|E_n\rho_n(X) - EW\left(\frac{\sigma Z}{\sqrt{1+\sigma^2}}\right)\right| \leq E_n\left|\rho_n(X) - EW\left(\frac{\sigma Z}{\sqrt{1+\sigma^2}}\right)\right| \\ \leq ME_n\|F_{X,n}-G_{X,n}\|.$$

The lemma follows because this expectation tends to zero by Theorem 16.23. □

Theorem 16.25. *Suppose our families are LAN at θ_0 in the interior of Ω and that $\hat\theta$ satisfies* (16.5)*, and let W be a bowl-shaped function. Then for any sequence of estimators δ_n,*

$$\lim_{b\to\infty}\lim_{c\to\infty}\liminf_{n\to\infty}\sup_{\|\theta-\theta_0\|\leq c/\sqrt{n}} E_\theta \min\big\{b, W\big(\sqrt{n}(\delta_n-\theta)\big)\big\} \geq EW(Z),$$

where $Z\sim N(0,J^{-1})$. The asymptotic lower bound here is achieved if $\delta_n=\hat\theta_n$.

Proof. Let $\pi_n = N(\theta_0, \sigma^2J^{-1}/n)$, the prior distribution for Θ in Lemma 16.24, and note that $\theta_0+\sigma Z/\sqrt{n}\sim\pi_n$. Also, let $W_b=\min\{b,W\}$, a bounded bowl-shaped function. Then

$$\begin{aligned}
\inf_\delta & EW_b\big(\sqrt{n}(\delta-\Theta)\big) \\
&\leq EW_b\big(\sqrt{n}(\delta_n-\Theta)\big) \\
&= \int E_\theta W_b\big(\sqrt{n}(\delta_n-\theta)\big)\,d\pi_n(\theta) \\
&\leq \sup_{\|\theta-\theta_0\|\leq c/\sqrt{n}} E_\theta W_b\big(\sqrt{n}(\delta_n-\theta)\big)\pi_n\big(\{\theta:\|\theta-\theta_0\|\leq c/\sqrt{n}\}\big) \\
&\qquad + b\pi_n\big(\{\theta:\|\theta-\theta_0\|>c/\sqrt{n}\}\big).
\end{aligned}$$

But

$$\pi_n\big(\{\theta:\|\theta-\theta_0\|\leq c/\sqrt{n}\}\big) = P\big(\|Z\|\leq c/\sigma\big).$$

Solving the inequality to bound the supremum and taking the lim inf as $n \to \infty$, using Lemma 16.24,

$$\liminf_{n\to\infty} \sup_{\|\theta-\theta_0\|\le c/\sqrt{n}} E_\theta W_b\big(\sqrt{n}(\delta_n-\theta)\big) \ge \frac{EW_b\big(\sigma Z/\sqrt{1+\sigma^2}\big) - bP\big(\|Z\|>c/\sigma\big)}{P\big(\|Z\|\le c/\sigma\big)}.$$

Letting $c \to \infty$ the denominator on the right-hand side tends to one, leaving a lower bound of $EW_b\big(\sigma Z/\sqrt{1+\sigma^2}\big)$. Because this lower bound must hold for σ arbitrarily large,

$$\lim_{c\to\infty} \liminf_{n\to\infty} \sup_{\|\theta-\theta_0\|\le c/\sqrt{n}} E_\theta W_b\big(\sqrt{n}(\delta_n-\theta)\big) \ge EW_b(Z).$$

The first part of the theorem now follows because $EW_b(Z) \to EW(Z)$ as $b \to \infty$ by monotone convergence. The second part that the asymptotic bound is achieved if $\delta_n = \hat\theta_n$ holds because $\sqrt{n}(\hat\theta_n - \theta) \Rightarrow N(0, J^{-1})$ uniformly over $\|\theta-\theta_0\| \le c/\sqrt{n}$. Using contiguity, this follows from (16.5) and normal approximation for the distributions of S_n mentioned in Remark 16.13. ☐

In addition to the local risk optimality of $\hat\theta$ one can also argue that $\hat\theta$ is asymptotically sufficient, as described in the next result. For a proof see Le Cam and Yang (2000).

Theorem 16.26. *Suppose the families $\mathcal{P}_n$ are locally asymptotically normal at every θ and that estimators $\hat\theta_n$ satisfy* (16.5). *Then $\hat\theta_n$ is asymptotically sufficient. Specifically, there are other families $\mathcal{Q}_n = \{Q_{\theta,n} : \theta \in \Omega\}$ such that:*

1. *Statistic $\hat\theta_n$ is (exactly) sufficient for $\mathcal{Q}_n$.*
2. *For every $b > 0$ and all $\theta \in \Omega$,*

$$\sup_{|\omega-\theta|\le b/\sqrt{n}} \|Q_{\omega,n} - P_{\omega,n}\| \to 0$$

as $n \to \infty$.

For a more complete discussion of asymptotic methods in statistics, see van der Vaart (1998), Le Cam and Yang (2000), or Le Cam (1986).

16.7 Problems

1. Consider a regression model in which $Y_i = x_i\beta + \epsilon_i$, $i = 1, 2, \dots,$ with the ϵ_i i.i.d. from $N(0, \sigma^2)$, and assume that $\sum_{i=1}^\infty x_i^2 < \infty$. Let Q_n denote the joint distribution of $Y_1, \dots, Y_n$ if $\beta = \beta_0$, and let $\tilde Q_n$ denote the joint distribution if $\beta = \beta_1$.

a) Show that the distributions Q_n and $\tilde{Q}_n$ are mutually contiguous.
b) Let L_n denote the likelihood ratio $d\tilde{Q}_n/dQ_n$. Find limiting distributions for L_n when $\beta = \beta_0$ and when $\beta = \beta_1$. Are the limiting distributions the same?

2. Let $X_1, X_2, \dots$ be i.i.d. from a uniform distribution on $(0, \theta)$. Let Q_n denote the joint distribution for $X_1, \dots, X_n$ when $\theta = 1$, and let $\tilde{Q}_n$ denote the joint distribution when $\theta = 1 + 1/n^p$ with p a fixed positive constant. For which values of p are Q_n and $\tilde{Q}_n$ mutually contiguous?
3. Prove the second assertion of Proposition 16.5: If the distributions for $\tilde{X}_n$ are contiguous to those for X_n, and if $T_n(X_n) = O_p(1)$, then $T_n(\tilde{X}_n) = O_p(1)$.
4. Let X and Y be random vectors with distributions P_X and P_Y. If h is a one-to-one function, show that

$$\|P_X - P_Y\| = \|P_{h(X)} - P_{h(Y)}\|.$$

In particular, if X and Y are random variables and $a \neq 0$,

$$\|P_X - P_Y\| = \|P_{aX+b} - P_{aY+b}\|.$$

5. Show that $X_n \xrightarrow{p} 0$ if and only if $E\min\bigl\{1, |X_n|\bigr\} \to 0$.
6. Define $g(x) = \min\{1, |x|\}$ and let $Z = E\bigl[|Y| \bigm| X\bigr]$. Show that $Eg(Z) \geq Eg(Y)$. Use this and the result from Problem 16.5 to show that $L(\Theta, X) \xrightarrow{p} 1$ when (16.13) holds.
7. Let Y_n be integrable random variables. Show that if $E\bigl[|Y_n| \bigm| Z_n\bigr] \xrightarrow{p} 0$, then $Y_n \xrightarrow{p} 0$.